Simulation of dust settling and planetesimal formation in protoplanetary discs

Yuanyuan Tan

Abstract— **Dust settling in protoplanetary discs is a key step toward planetesimal formation. This project develops a one-dimensional numerical model that combines first-order upwind advection with centred turbulent diffusion, validated through analytic tests and convergence studies (~3% numerical uncertainty). In the dense Solar-System-motivated disc adopted here, 1-μm grains settle toward the midplane only on Myr timescales and show little radial dependence when radii greater than 50AU. Turbulent mixing strongly suppresses this settling matching observed dust-layer thicknesses at 100 AU requires very weak turbulence $\alpha = 10^{-4}$. Under such conditions, micron grains cannot reach midplane enhancement within 1 Myr, implying that grain growth is essential. This is consistent with the size-scan results, which show that even substantially larger dusts cannot overcome turbulent mixing within 1 Myr in the dense disk model.**

Overall, both weak turbulence and sufficiently large dusts are required for producing the concentrations needed for planetesimal formation.

I. Introduction

After the formation of young stars, planets develop within protoplanetary discs, which are accretion discs composed of 99% hydrogen and helium gas and 1% solid particles (dust).

Planetesimals are the precursor stage of planets. Dust settles because the vertical component of stellar gravity pulls particles toward the midplane, while the gas pressure gradient provides an upward force that can balance gravity. Therefore, planetesimal formation requires the solid material to become sufficiently concentrated, typically with a local dust–gas ratio equals 1 or greater. In this case, streaming instability leads to formation of planetesimals.

In these processes, the vertical settling driven by gravity can be disturbed by turbulent diffusion. The aim of this project is to quantify these processes using a controlled 1D vertical model of the disc to simulate dust settling. The equation is solved using an explicit finite-difference scheme: an upwind method is used for advection, and a centred scheme for turbulence-driven diffusion.

The report aims to answer four questions.

First, the numerical methods must be validated through convergence tests and an examination of numerical diffusion in the upwind scheme. Second, what is the timescale for small grains settling in the absence of turbulence, and how does this timescale vary with radius? Third, how does turbulence prevent dust from fully settling into the midplane. Finally, as dusts grow, which sizes can overcome turbulence and allow planetesimal formation to occur? All these questions are investigated using disc parameters chosen to resemble those of the early Solar System. In this case, we can understand the limitation about dust in the outer disc can reach the conditions required for planetesimal formation.

II. Methods and Numerical Setup

A. Upwind and Diffusion Method

The vertical dust settling obeys an advection-diffusion equation, which is present as follows:

$$\frac{\partial \rho_d}{\partial t} + \frac{\partial}{\partial z}(u_z \rho_d) = D\,\frac{\partial}{\partial z}\left[\rho_g \frac{\partial}{\partial z}\left(\frac{\rho_d}{\rho_g}\right)\right] \quad (1)$$

Where $\rho_d(z)$ is the dust density, $\rho_g(z)$ is the gas density, effective diffusion constant $D = \alpha \frac{c_s^2}{\Omega}$, where α is a dimensionless constant, Ω is circular Keplerian orbital frequency, c_s is the sound speed.

The terminal velocity approximation $u_z = -\Omega^2 z\, t_s(z)$ is used, valid for small grains in a thin disc. Here $t_s(z)$ here is the settling timescale proportional to dust radius. The gas is assumed to be in vertical hydrostatic equilibrium, giving the

fixed Gaussian distribution.

$$\rho_g(z) = \rho_{g0} \exp\left(-\frac{z^2}{2H^2}\right) \quad (2)$$

The change in dust density depends on the advective flux with negative velocity u_z and on diffusion caused by local turbulence on the righthand side of the equation.

The advection term is discretized using a first-order upwind flux. This method is monotonic, ensures stability and non-oscillatory behavior near steep gradients, and gives a fully explicit time integration that keeps the computational cost low.

Although upwind schemes introduce numerical diffusion, the convergence test verify that it does not affect physical conclusions when an appropriate resolution is used.

The turbulent diffusion term is discretized using a second-order centred finite-difference approximation, which provides a consistent and symmetric treatment of diffusion. This scheme is also compatible with the explicit timestep constraint imposed by the diffusive CFL condition.

Before discretization, the equation is rewritten in non-dimensional form using the disc scale height $H$and orbital frequency Ω. This transformation removes the large variations of scales of the dimensional quantities, improves the numerical stability of the explicit scheme. The normalized function is shown in equation (3).

$$\frac{\partial \widetilde{\rho_d}}{\partial \tilde{t}} = -\frac{\partial}{\partial \tilde{z}}(\widetilde{u_z}\,\widetilde{\rho_d}) + \alpha \frac{\partial}{\partial \tilde{z}}\left[\widetilde{\rho_g}\frac{\partial}{\partial \tilde{z}}\left(\frac{\widetilde{\rho_d}}{\widetilde{\rho_g}}\right)\right] \quad (3)$$

$$where, \tilde{z} = \frac{z}{H}, \tilde{t} = \Omega t, \widetilde{\rho_d} = \frac{\rho_d}{\rho_{g0}}, \widetilde{u_z} = \frac{u_z}{H\Omega}$$

B. Boundary conditions

Two physical boundary conditions are applied to the one-dimensional vertical domain.

At the midplane $z = 0$, the dust distribution is assumed to be symmetric. Therefore, the vertical velocity should satisfy $u_z = 0$, which prevents any transport of dust across the midplane that would cause asymmetry.

At the upper boundary $z = z_{\max}$, the disc is assumed not to receive dust inflow from vacuum, outside of the computational domain. Accordingly,$\rho_d(z_{\max} + 1) = 0$ is imposed. Although this resembles an open boundary condition that would allow dust to flow out, in practice dust does not leave the domain because its vertical velocity is directed toward the midplane.

C. Choice of Timestep and Optimization

The finite-difference grid introduces an artificial timescale, and the timestep must stay below that scale to remain stable. Therefore, the timestep Δt must satisfy the stability constraints of both advection and diffusion:

$$\widetilde{\Delta t} \le \min\left(C_{adv}\frac{\tilde{h}}{\max|\widetilde{u_z}|}, C_{diff}\frac{\widetilde{h^2}}{\alpha\,\widetilde{\rho_g}}\right) (4)$$

The constants $C_{\mathrm{adv}}$and $C_{\mathrm{diff}}$are safety factors between 0 and 1 chosen manually. The first term in equation (4) comes from the advection CFL limit. The second term comes from the diffusion stability condition. The timestep is the minimum of the two, and the detailed choice of these constants is discussed later.

In practice, the settling velocity $\tilde{u}_z$grows with height because it is proportional to z, and even faster than the speed of sound, which is non-physical because the terminal-velocity approximation assumes subsonic dust–gas relative motion. This makes the advection limit very strict near the edge of disc, causing the timestep to become extremely small, near 5×10^{-6}yr.

However, the expression for $\tilde{u}_z$is not reliable in the very low-density upper layers, where the physical assumptions behind the terminal velocity formula break down. Using the unmodified value there would increase the computational cost without improving the accuracy. Therefore, a simple cap $|\tilde{u}_z| \approx 0.9C_s$ is applied to avoid extremely small timesteps the final timestep is thus a balance between stability and efficiency.

Also, to improve computational efficiency, vectorized operations are used instead of for loops.

III. Code Verification

This section verifies the correctness of the code and establishes appropriate numerical conditions for the solver.

A. Advection test

A simple one-dimensional test with a known analytic solution was used to verify the correctness of the advection step.

The initial condition is a smooth Gaussian pulse on a periodic domain. For a constant advection velocity, the exact solution is a shift of the initial

points. Figure 1shows the numerical solution at $t = 0.5$for four different spatial resolutions ($N_x = 50, 150, 200, 500$), together with the analytic profile.

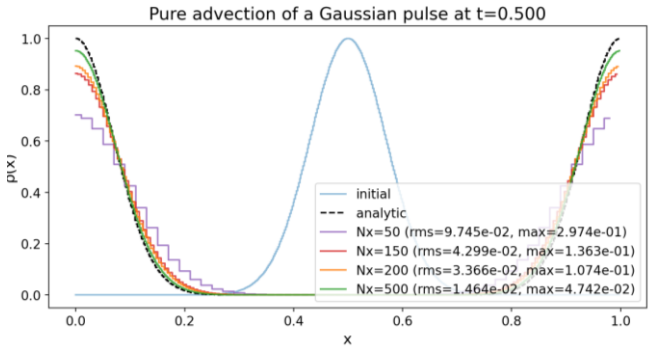


Fig. 1. Pure 1D advection equation test result. Where "rms" is the root-mean-square error between the numerical and analytic function, and "max" is the maximum deviation.

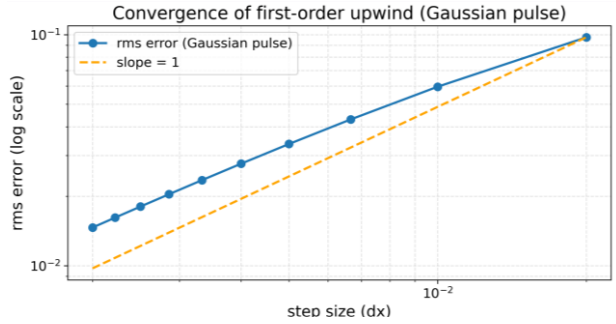


Fig. 2. Convergence of first-order upwind method for pure advection.

Several expected behaviors of a first-order upwind method appear in the figure. First, the simulation matches the analytic result but exhibits numerical diffusion. Second, the amount of numerical diffusion decreases as the grid is refined. Both decrease systematically with increasing resolution, confirming that the method is convergent.

Figure 2 shows the grid resolution study for pure advection using a Gaussian pulse. The rms error decreases approximately linearly as the grid spacing (step size) decreases on a log-log plot, with a measured slope close to 1. The numerical errors lie slightly above the ideal slope–1 line due to the inherent numerical diffusion of the upwind discretization, but the overall order of accuracy is unchanged. This confirms that the upwind method is first–order accurate for smooth solutions.

The test therefore verifies both the correctness of the advection method and the characteristic numerical diffusion of the upwind discretization.

B. Numerical Robustness and Parameter Choices

To check that the numerical solution does not depend on the choice of parameters, several robust tests were performed, and the test result is presented in table.1.

Nz	Final ratio ρd/ρg	t_settle (Myrs)
150	1.491	1.081000
200	1.985	1.017000
250	2.477	0.994000
300	2.966	0.983000
CFL factor	**Final ratio ρd/ρg**	**t_ settle (Myrs)**
0.30	1.9849	1.016593
0.50	1.9849	1.016804
0.70	1.9850	1.016593
0.90	1.9851	1.016910
V cap factor	**Final ratio ρd/ρg**	**t_ settle (Myrs)**
0.30	1.9851	1.016910
0.50	1.9851	1.016910
0.70	1.9851	1.016910

Table. 1. Numerical Robustness Test.
the tests focus on three aspects: grid resolution along z direction, timestep scaling, and the velocity–cap parameter used to avoid extremely small steps and non-physical speed. "Final ratio ρd/ρg" is the mid–plane dust–gas ratio at the final time. "t_settle" is the first time in unit of Myr when dust–gas ratio becomes greater than 1.

The resolution test varies the number of vertical grid points while keeping all other numerical parameters fixed. The trend acts as expected, higher resolution gives a stronger midplane concentration. The change in t_{settle}becomes small for $Nz \geq 200$. Between $Nz = 200$ and Nz = 300 being only ≈3%, implying that the result has converged. The simulations presented in later sections therefore adopt $Nz = 250$, since it is a balance between accuracy and computational cost.

To test sensitivity to the timestep, the baseline CFL–limited timestep was scaled by CFL factor. The results show that this change does not affect the dust–gas ratio, provided the timestep satisfies the constraints described in Section 2C. and t_{settle}varies by less than 0.03%. Therefore, the diffusion–number constraints C_{diff} and advection constraints C_{adv} can be 0.7.

A soft velocity cap $|\tilde{u}_z| \approx 0.9C_s$ is applied in the initial settling velocity to avoid unphysical speeds far above the sound speed, and to reduce the number of timesteps, the test result confirms that when the dust size is $1\mu m$,(the case used in result section A-B of the result discussion) the velocity cap is never triggered.

To avoid nonphysical supersonic settling speeds for large dust in the result section C (eg.1cm), the terminal velocity may exceed the sound speed, a

velocity cap was introduced. In this case, the cap affects the absolute value of settling time. However, the particle size required to overcome turbulence is stable to within an order of magnitude, so the qualitative conclusions of the previous section are unaffected.

From all tests, the combined evidence shows that the solver is robust, and that the parameter choices used in the following sections do not bias physical interpretation.

IV. Results & Discussion

All simulations use the solar–nebula disk conditions provided in the project brief. The gas density is as follows:

$$\rho_{g0}(R) \approx 5 \times 10^{-7} \left(\frac{R}{1\,\mathrm{AU}}\right)^{-3} \ \mathrm{kg\,m^{-3}}$$

with temperature $T \propto R^{-1/2}$, sound speed $c_s \propto R^{-1/4}$, and Keplerian frequency $\Omega \propto R^{-3/2}$. Dust is initialized with a vertically uniform dust–to–gas ratio of 0.01 and particle density of $1\times10^{3}\mathrm{kgm^{-3}}$. Unless otherwise stated, the grain size is $a = 1\ \mu\mathrm{m}$. All simulations evolved up to a maximum time of 1 Myr, which is a representative outer–disk lifetime and sets a physically meaningful upper limit for the enhancement of midplane.

As we discussed in previous section, The diffusion–number constraints C_{diff} and advection constraints C_{adv} ARE 0.7, and the number of vertical grid points Nz =250. This gives a numerical uncertainty of approximately 3%.

These choices represent a dense version of the solar nebula and fix all dimensional coefficients in the settling and diffusion terms. The results section starts from the non-turbulence case. It first simulated the vertical settling of 1 μm size of dust without turbulent diffusion. The settling time $t_{\mathrm{settle}}$was defined as the first moment when the midplane dust–gas ratio exceeds 1.

A. No Turbulence Dust settling

To quantify how efficiently micron–sized grains can concentrate toward the midplane in the absence of turbulence. The 1D vertical settling equation is solved.

Across 20–100 AU, micron grains settle slowly. All radii require roughly $1-2$Myr to reach $\rho_d/\rho_g > 1$at the midplane, with only a weak radial trend. At 20 AU the settling time is close to $\sim 1.8-2.0$Myr, decreasing toward about 1.0Myr at 100 AU (0.994 Myr in the simulation), which lies just within typical protoplanetary disc lifetimes. The spread between radii is substantially larger than the 3% numerical uncertainty from resolution and CFL tests, so the trend is robust.

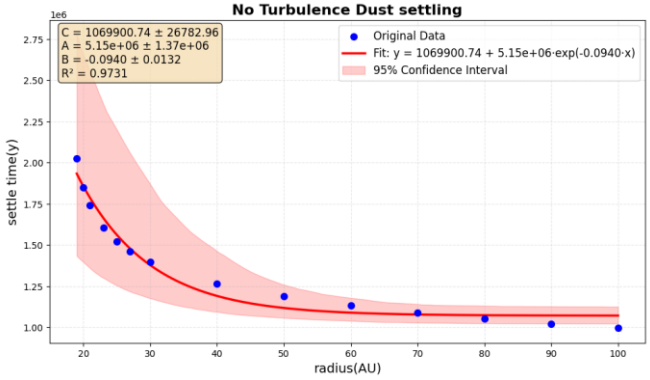


Fig. 3. Settling time t_settle for 1 μm dust grains in the absence of turbulence. The solver was run across 20–100 AU. Fitting Uncertainty is presented at the figure.

Figure 3 shows that the fitted relation is well described by an exponential with a constant offset. This is a very weak exponential and can be ignored when radius is greater than 100AU. The radial dependence becomes negligible when calculating settle time in this range. In the outer disk the gas becomes thinner, but it also becomes colder and less gravitationally bound. These effects act in opposite directions on the settling speed and produce a near balance.

In the dense solar–nebula disk adopted here, micron grains remain tightly coupled to the gas and settle inefficiently. Even without turbulence, reaching midplane enhancement requires timescales comparable to the upper end of typical protoplanetary disk lifetimes about$1-3$ Myr. Since settling alone already operates on Myr timescales, even a modest amount of mixing will matter, so we next examine how turbulence modifies the vertical structure.

B. Turbulent diffusion and with different diffusion numbers 100 AU

In the turbulent case, each model evolves for 10^{7}yr, which is an order of magnitude longer than the settling time without turbulence. At this time, the vertical dust profile is close to its steady shape, and the dust-layer thickness can be measured.

Figure 4 shows the resulting dust–gas ratio profiles at 100 AU for four different diffusion

strengths. It shows that increasing the diffusion parameter broadens the dust layer in steady state.

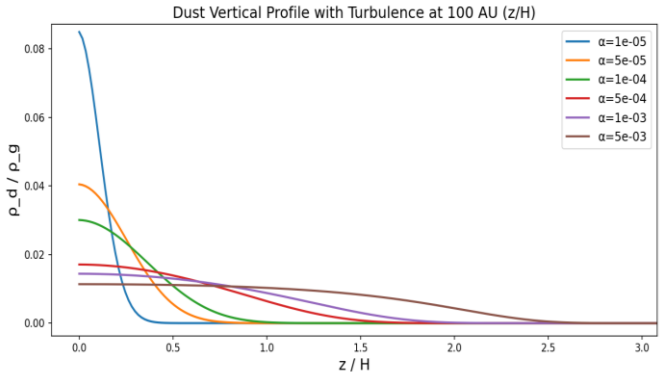


Fig. 4. Dust–gas ratio at 100 AU in steady state for different turbulent diffusion strengths. α denotes the diffusion parameter. The vertical coordinate z is scaled by the gas scale height H; in physical units, $z/H \approx 0.1R$. Only the positive-z side of the profile is shown.

When diffusion is weak ($\alpha = 10^{-5}$), the dust remains tightly concentrated near the mid-plane, while stronger diffusion ($\alpha = 5 \times 10^{-3}$) produces a noticeably thicker and more extended layer. This behavior reflects the fact that random turbulent motions continuously mix dust upward, diffuse away from mid-plane, slowing the net settling speed and producing a broader vertical layer.

Observations of protoplanetary discs in Taurus star cluster show that, at $R \approx 100$AU, this dust-layer thickness is roughly $z \approx 0.1R$[1]. Using this as a reference, the simulated profiles are compared with different turbulent diffusion strengths. Simulations with $\alpha = 10^{-4}$to 5×10^{-4}produce dust distributions that extend to similar scaled heights.

Recent observational constraints from a sample of 33 protoplanetary disks indicate that in most outer discs the vertical turbulence is weak, satisfying $\alpha \lesssim 10^{-3}$ under the assumption of fragmentation-limited dust sizes.[2] This observationally motivated range in the order of $\alpha \approx 2 \times 10^{-4}$ is then applied in the subsequent simulations about settling times. Again, although there is some numerical uncertainty,the qualitative trend—that stronger diffusion produces a broader dust layer—is unchanged across all tests.

Also, notice that $1\mu m$ size cannot evolve to the settle condition, it with stay a stable equilibrium with a board or narrow dust layers. This confirms the classical expectation that small grains remain too tightly coupled to the gas to participate in direct gravitational collapse. This suggests that larger dust grains are required to form planetesimals within the limited 1 Myr timescale.

C. *Dust growth and critical size*

Question is therefore whether dust growth can compensate for turbulent mixing and allow the midplane dust–to–gas ratio to exceed unity within the disk lifetime.

To test this, particle sizes over several orders of magnitude are scanned and measured whether midplane enhancement occurs within 1 Myr.

Here the turbulent parameter α is fixed to the representative value inferred in Section B ($\alpha = 2 \times 10^{-4}$) and dust growth at radii between 30 and 100 AU is evaluated. Particle radius is varied from 10 to $10^6 \mu$m, and each simulation is stopped once the midplane dust–to–gas ratio exceeds unity or when the 1 Myr limit is reached.

Across the entire scanned size range (up to1cm), no particle population achieves $\rho_d/\rho_g > 1$within 1 Myr. Larger grains do settle faster, but turbulent diffusion continues to redistribute material away from the midplane, and the vertical drift never becomes strong enough to overcome mixing. The dense gas plays a central role here. Higher gas density strengthens the aerodynamic drag on particles. As a result, the settling speed increases much more slowly with size than in lower–density disk models.

A secondary effect arises from the open vertical boundaries of the diffusion term. The open vertical boundary allows material to leave the domain, but because dust always drifts toward the midplane, the outflow is negligible and only gives a tiny effect on results.

For the dust sizes above $10^6 \mu$m(order-metre scale), the velocity assumptions underlying this vertical–settling model are no longer valid. The terminal velocity formula produces non-physical speeds, and at least 20% of the computed velocities are limited by the cap. In this case, the results are no longer physically meaningful, and the full vertical equation of motion would need to be solved instead of assuming instantaneous balance between drag and gravity.

Overall, in the dense disk, vertical settling combined with isotropic turbulence is insufficient to produce midplane self-enrichment in this model. This limitation again reflects the behavior seen in the size scan simulations, where settling alone never produces midplane enhancement

within the available Myr timescale. If such a disk were representative of the early solar system, additional concentration mechanisms like pressure bumps, zonal flows, or streaming instability (which needs a high dust-gas ratio) would be required to trigger planetesimal formation.

By contrast, a lighter disk density may allow an efficient settling once particles reach mm–cm sizes, enabling midplane enhancement within a Myr. Equation for Minimum Mass Solar Nebula density is:

$$\rho_{g0}(R) \approx 1.4 \times 10^{-9} \left(\frac{R}{1\,\mathrm{AU}}\right)^{-11/4} \ \mathrm{kg\,m^{-3}}$$

Such efficient settling of mm–cm sized grains in a lower-density disk, is consistent with previous models of dust evolution and planetesimal formation.[3] Although this regime is not directly simulated here, the expected transition size should increase with radius, as weaker vertical gravity and larger scale height make settling progressively less efficient in the outer disk.

V. Conclusion

This project investigated vertical dust settling in a protoplanetary disc using a 1D advection–diffusion model. The simulations lead to three main findings.

First, micron-sized grains cannot achieve midplane enhancement within 1 Myr in a turbulent disc. Even in the absence of turbulence, 1 μm particles reach midplane densities above the gas only on Myr timescales, indicating that they cannot form planetesimals without additional processes.

Second, turbulence must be weak for efficient settling. Matching the observed dust-layer thickness at 100 AU requires $\alpha \approx 10^{\wedge -5}$–$5\times10^{-5}$ in the dense disc model prescribed by the project, significantly lower than typical values in lighter discs.

Third, dust growth is expected to play a key role in improving midplane enhancement. In the dense disk model used in this project, even very large grains fail to overcome turbulent mixing within 1 Myr, indicating that growth alone is insufficient under such conditions. This is fully consistent with the size-scan results, which show that no tested grain size achieves midplane enhancement within 1 Myr in the dense-disk model. However, theoretical models and observations of lighter disks suggest a different trend: once particles grow to mm–cm sizes, their settling speed can become comparable to, or exceed, the level of turbulent diffusion.

All results are robust, as shown by numerical convergence tests and parameter variations. The advection scheme behaves consistently under refinement, and the qualitative conclusions do not depend on grid resolution or timestep.

In summary, planetesimal formation requires both grain growth and weak turbulence. These trends are consistent with observational evidence of pebble-sized material in young systems and with theoretical expectations for dust evolution in protoplanetary discs.

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Acknowledgement

Generative AI (ChatGPT) was used for writing assistance and gramma editing.